

Power BI Sales Analytics Project Documentation

This document explains the complete Power BI Sales Analytics project workflow from data preparation to dashboard creation, visualization development, drill-through functionality, and tooltip implementation.

1. Dataset Overview

The project uses multiple CSV datasets to build an end-to-end sales analytics dashboard in Power BI.

- Customer Table
- Product Categories
- Product Subcategories
- Products
- Returns
- Sales 2015
- Sales 2016
- Sales 2017
- Territories

2. Data Cleaning & Transformation Process

1. Changed the regional settings to English (United States).
2. Changed the Birthdate column data type from Text to Date.
3. Replaced 'Mgmnt' with 'Management' in the Occupation column.
4. Removed the '\$' symbol from the Annual Income column.
5. Changed the Annual Income data type from Text to Whole Number.

3. Overview Dashboard Tab

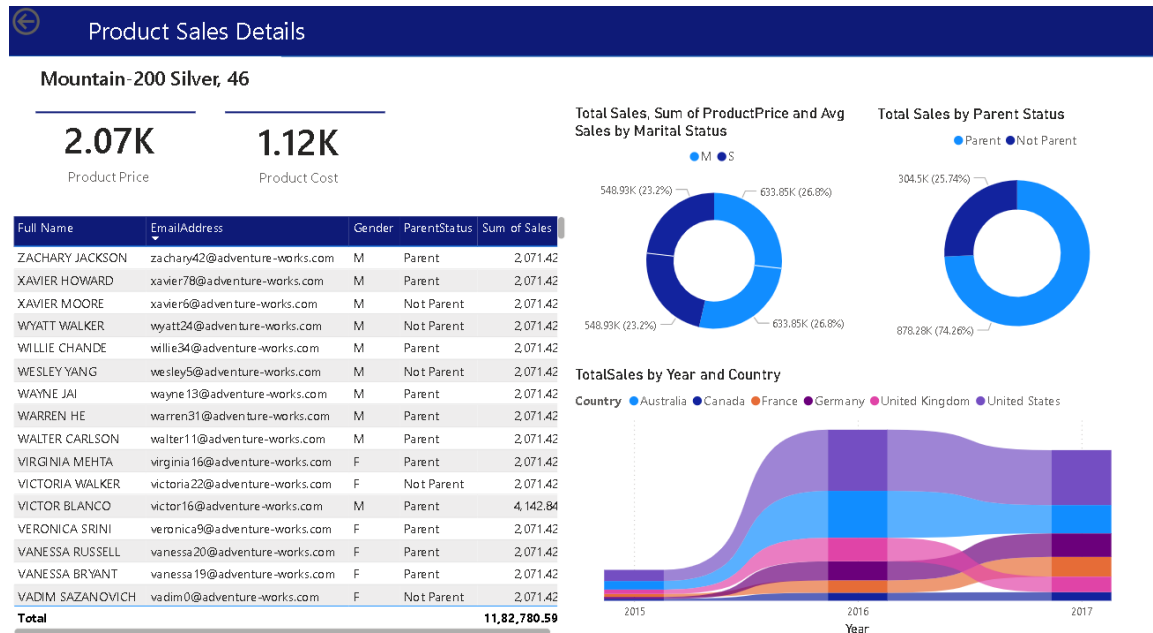
The Overview dashboard provides a high-level summary of the sales performance, order trends, product performance, and return analysis.



- KPI Cards created for Total Sales, Total Orders, Total Customers, and Return Percentage.
- Stacked Bar Chart created for Sales by Sub-Category and Status.
- Line Chart created for Sum of Sales by Date.
- KPI Cards created for Last Month Sales and Last Month Orders.
- Interactive table created with Product Name, Total Sales, Order Count, and Total Returns.
- Field Parameters implemented with single-select functionality.
- Dimension Selector implemented using Customer Name and Product Name parameters.
- Map visualization created for Total Sales by Country.
- Slicers added for Category Name, Gender, Region, Email Address, and Year Quarter.

4. Product Details Dashboard Tab

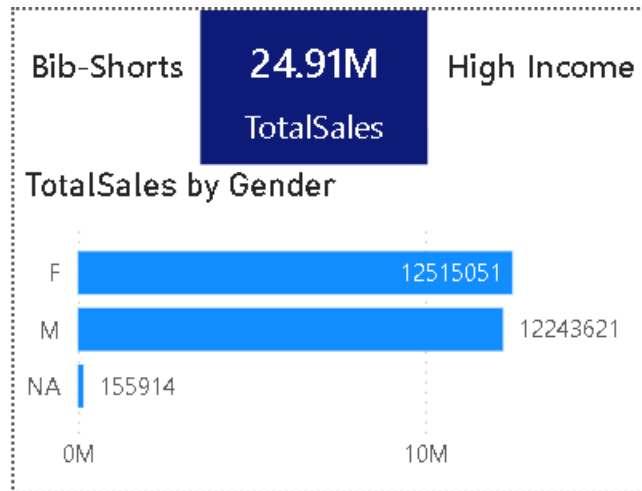
A drill-through functionality was created from the Overview dashboard table to access detailed product-level analysis.



- Drill-through navigation enabled from the Overview table to Product Details page.
- KPI Cards created for Product Price and Product Cost.
- Ribbon Chart created for Total Sales by Year and Country.
- Pie Chart created for Total Sales by Marital Status.
- Pie Chart created for Total Sales by Parent Status.
- Detailed customer table created with Full Name, Email Address, Gender, Parent Status, and Sum of Sales.

5. Tooltip Page

A custom tooltip page was designed to provide additional insights when users hover over visual elements in the Overview dashboard.



- Added Sub-Category card.
- Added Income Status card.
- Added Total Sales card.
- Added Bar Chart for Total Sales by Gender.
- Enabled tooltip visibility on the Sales by Sub-Category and Status chart in the Overview tab.

6. Complete Project Workflow

6. Imported all CSV datasets into Power BI.
7. Performed data cleaning and transformation using Power Query.
8. Merged yearly sales files into a consolidated dataset.
9. Created relationships between tables in the data model.
10. Developed calculated measures and KPIs using DAX.
11. Designed Overview dashboard visuals and slicers.
12. Implemented drill-through navigation for product-level analysis.
13. Created a custom tooltip page for enhanced user interaction.
14. Validated data accuracy and dashboard performance.
15. Published the final interactive dashboard.

7. Recommendations

a. Focus on High-Performing Product Categories

The dashboard analysis shows that **Road Bikes and Mountain Bikes** are the top-performing product categories. The business should increase inventory levels, strengthen marketing campaigns, and run promotional activities for these products to maximize revenue and market demand.

b. Improve Low-Performing Products

Products such as **Socks, Cleaners, Vests, and Caps** are generating lower sales compared to other categories. To improve performance, the business can:

- Offer discounts and bundle deals
- Reposition these products through targeted marketing campaigns
- Evaluate product demand and discontinue underperforming items if required

c. Reduce Product Return Rates

The most returned products are **Water Bottles, Patch Kits, and Mountain Tire Tubes**. These products should be analyzed further to identify possible issues such as:

- Product quality concerns
- Incorrect or unclear product descriptions
- Packaging or delivery-related damages

Reducing return rates can improve customer satisfaction and increase overall profitability.

d. Expand Focus on High-Revenue Regions

The dashboard indicates that the **USA, Australia, France, Germany, and the UK** generate the highest sales revenue. The business should:

- Expand operations in these high-performing regions

- Increase advertising and promotional budgets
- Develop region-specific marketing strategies

e. Enhance Customer Retention Strategies

By analyzing customer-level sales and order data, the business can identify repeat customers and high-value buyers. Recommended strategies include:

- Implementing customer loyalty programs
- Offering exclusive deals for returning customers
- Running personalized email and communication campaigns

These initiatives can help improve customer retention and long-term customer value.